

Daniel-John Diala

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EDUCATION

Monmouth University, West Long Branch, NJ

Expected May 2029

Bachelor of Science in Software Engineering · Honors School

GPA: 4.0/4.0

Honors: Dean's List (Fall 2025, Spring 2026); Academic Excellence Scholarship; Shirley Family Scholarship

Coursework: Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Calculus I & II

SKILLS & CERTIFICATIONS

Languages: Python, Java, TypeScript, JavaScript, SQL, OCaml, HTML/CSS

ML & Data: PyTorch, spaCy, NLP, federated learning, differential privacy, Plotly

Web & Infra: FastAPI, Flask, React, Next.js, PostgreSQL, Redis, AWS (EC2), Docker, Git, GitHub Actions, pytest

Certification: Claude Certified Architect – Foundations, Anthropic (Jul 2026)

EXPERIENCE

Undergraduate Research Assistant | Monmouth University

May 2026 – Present

- Modernized GAreL, an OCaml bidirectional type checker (from Qu et al., ICFP 2019) that analyzes program cost statically
- Fixed drift in a 36-program test corpus, stress-tested the checker, and developed one of two independent solution paths
- Currently evaluating CHC- and SMT-solver-based alternatives to scale the analysis
- Built an educational web app, hosted on AWS, that visualizes constraint generation to teach bidirectional type checking

AI Literacy Teaching Assistant | Monmouth University

Sep 2026 – Present

- Help students set up agentic AI systems and use AI tools effectively and ethically
- Built a Next.js/TypeScript exam-prep web app with 180 practice questions and timed mock exams for Anthropic's Claude Certified Architect – Foundations exam, deployed on AWS EC2

PROJECTS

Pulse-Net: Federated Pandemic Early-Warning System | Python, PyTorch, FastAPI, PostgreSQL, React

Jan 2026 – Present

- Built a federated learning pipeline where 5 simulated hospitals train PyTorch models locally and send signed, differentially private ($\epsilon = 1.0$) updates, merged with a norm-clipped coordinate-wise median to resist poisoning
- Designed a transformer classifier with supervised contrastive loss that flags novel pathogens by embedding distance; 83.6% validation accuracy across 16 classes on synthetic data
- Built encrypted aggregation with CKKS, Shamir k-of-n key escrow, zero-knowledge range proofs, and a Merkle audit log
- Built a React dashboard with a live outbreak map, WebSocket alerts, and SEIR spread models; 1,500+ pytest tests run in CI

Sentiment Analysis Tool | Python, spaCy, Flask, SQLite, Streamlit, Plotly

Dec 2025 – Jan 2026

- Built an NLP engine that classifies reviews as positive, negative, neutral, or mixed using a 6,800-term graded lexicon with domain variants, negation handling, and emoji sarcasm cues
- Added aspect-based analysis and a Flask REST API (15+ endpoints) feeding a Streamlit trend dashboard
- Cut runtime for 500 reviews from 17s to 5s (3.4×) via batched spaCy processing, lemma caching, and bulk SQLite inserts

ACTIVITIES

IEEE/ACM Student Chapter, University & External Relations Committee Chair

Mar 2026 – Present

CyberHawks: National Cyber League (Spring 2026), team top 4%, individual top 14%

Sep 2025 – Present